

AGRICULTURAL TECHNOLOGY COMPETITION

PROJECT PRESENTATION



PROJECT NAME: MOONHARVEST - PRECISION AGRICULTURE
MONITORING SYSTEM

TOPIC TITLE: CROSS-PLATFORM GROUND CONTROL STATION WITH
REAL-TIME COMPUTER VISION

TEAM NAME: EFRISA

TEAM ID: 783316

APPLICATION ID: 4800877



TEAM STRUCTURE



TEAM ADVISOR



TEAM CAPTAIN



MEMBER 1
ELECTRICAL DIVISION
3RD INDUSTRIAL
ELECTRONICS ENGINEERING



MEMBER 2
MEVHANICAL DIVISION
1ST ELECTRONICS
ENGINEERING



MEMBER 3
ELECTRICAL DIVISION
1ST ELECTRONICS
ENGINEERING



MEMBER 4
ELECTRICAL DIVISION
2ND ELECTRONICS
ENGINEERING



MEMBER 5
NON-TECHNICAL DIVISION
1ST TELECOMMUNICATIONS
ENGINEERING

TEAM PROFILE

PROJECT SUMMARY, SCOPE
AND TARGET AUDIENCE

PROJECT OBJECTIVE
AND SOCIAL BENEFIT

DEFINITION AND SIGNIFICANCE
OF THE PROBLEM

PROPOSED SOLUTION

METHOD

PROTOTYPE / DESIGN
AND FUNCTIONING

COMPARISON WITH
EXISTING SOLUTIONS

COMMERCIALISATION
POTENTIAL AND SUSTAINABILITY

PROJECT TIMELINE

MoonHarvest - Precision Agriculture Monitoring System

MAIN IDEA

An affordable, offline ground control station that turns any low-cost UAV into a real-time crop-health scanner using on-board AI.

Directly supports digital agriculture, sustainable production, and accessible smart farming.

Target Audience

Smallholder (1–10 ha) and medium (10–50 ha) farmers, cooperatives, and agronomy consultants in emerging markets.

Solution Approach

One cross-platform app (Windows / Linux / Android) streams drone video, runs YOLO locally, overlays vegetation health, and plans flights without internet.

Expected Impact

Early disease detection, 15–25% less pesticide (target), safer harvests for farmers who can't afford expensive drones.

Theme Alignment

Affordable smart farming for smallholders using an Android phone & a cheap drone, with no internet.

TEAM PROFILE

PROJECT SUMMARY, SCOPE AND TARGET AUDIENCE

PROJECT OBJECTIVE AND SOCIAL BENEFIT

DEFINITION AND SIGNIFICANCE OF THE PROBLEM

PROPOSED SOLUTION

METHOD

PROTOTYPE / DESIGN AND FUNCTIONING

COMPARISON WITH EXISTING SOLUTIONS

COMMERCIALISATION POTENTIAL AND SUSTAINABILITY

PROJECT TIMELINE

Project Objective & Social Benefit

OBJECTIVE

Democratize precision agriculture by delivering professional-grade aerial crop monitoring on affordable hardware that works fully offline.



Rationale

Existing precision-agriculture tools can cost USD 10k–30k and often require connectivity, making them inaccessible to many smallholder farmers.



Core Problem

Crop stress is often detected too late, while existing monitoring tools remain costly and connectivity-dependent.



Social Contribution

Stronger food security, less chemical runoff, and higher smallholder income across emerging markets.



Farmer Benefit

An affordable, offline tool brings early detection, targeted spraying, less labor, and higher yields.

TEAM PROFILE

PROJECT SUMMARY, SCOPE
AND TARGET AUDIENCE

PROJECT OBJECTIVE
AND SOCIAL BENEFIT

DEFINITION AND SIGNIFICANCE
OF THE PROBLEM

PROPOSED SOLUTION

METHOD

PROTOTYPE / DESIGN
AND FUNCTIONING

COMPARISON WITH
EXISTING SOLUTIONS

COMMERCIALISATION
POTENTIAL AND SUSTAINABILITY

PROJECT TIMELINE

The Problem in Five Dimensions

40%

PRODUCTION

Of global crop yield lost to pests & disease each year (USD 220B, FAO).

4–6 hrs

LABOR

To manually scout 10 ha done infrequently, so problems are caught late.

15–25%

ENVIRONMENT

Pesticide over-applied via blanket spraying, polluting soil & water.

\$220M

LOCAL ECONOMY

Estimated annual crop-disease loss in Indonesia (Panotra, 2025).

\$10k–30k

ACCESS GAP

Cost of existing precision-ag platforms and they still need connectivity.

SIGNIFICANCE

MoonHarvest directly addresses these five gaps by enabling earlier detection, automated scouting, targeted treatment, lower losses, and offline access at a lower cost.

TEAM PROFILE

PROJECT SUMMARY, SCOPE AND TARGET AUDIENCE

PROJECT OBJECTIVE AND SOCIAL BENEFIT

DEFINITION AND SIGNIFICANCE OF THE PROBLEM

PROPOSED SOLUTION

METHOD

PROTOTYPE / DESIGN AND FUNCTIONING

COMPARISON WITH EXISTING SOLUTIONS

COMMERCIALISATION POTENTIAL AND SUSTAINABILITY

PROJECT TIMELINE

Who Is Affected and Why Now?

FROM LATE DETECTION TO LOST INCOME

- 1 Manual, infrequent field scouting
- 2 Crop stress & disease detected too late
- 3 Problems spread across the field
- 4 Yield & quality losses
- 5 Lower income & weaker food security

MULTI-DIMENSIONAL EVIDENCE

80%

DEMOGRAPHIC

of the world's 570M farms are smallholders least equipped to absorb losses (FAO).

1/3

FOOD SUPPLY

of the world's food is grown by smallholders, so their losses hit supply directly (FAO).

50–60%

TEMPORAL TREND

projected rise in global food demand by 2050, raising the cost of every lost harvest (FAO).

\$220B

ECONOMIC

lost to pests & disease across world agriculture every single year.

TEAM PROFILE

PROJECT SUMMARY, SCOPE AND TARGET AUDIENCE

PROJECT OBJECTIVE AND SOCIAL BENEFIT

DEFINITION AND SIGNIFICANCE OF THE PROBLEM

PROPOSED SOLUTION

METHOD

PROTOTYPE / DESIGN AND FUNCTIONING

COMPARISON WITH EXISTING SOLUTIONS

COMMERCIALISATION POTENTIAL AND SUSTAINABILITY

PROJECT TIMELINE

How MoonHarvest Works

OPERATING LOGIC



Low-cost UAV



Live video +
MAVLink telemetry



YOLO / ONNX + Classify 4 states (Lush Green, Inconsistent Growth, Drought/Severe Stress, Bare Soil/Gap)

WHY IT IS EFFECTIVE

1

Fully offline & fast

Runs entirely on-device at >15 FPS on a standard laptop works in fields with no internet.

2

Earlier, targeted action

Detects stress before it is visible, enabling targeted spraying instead of blanket spraying (15–25% less chemicals).

3

Faster fieldwork

One automated flight maps 10 ha in minutes work that took 4–6 hours by hand.

DAILY USE A farmer or cooperative flies a 10–50 ha block, watches the live health map, and treats flagged zones the same day.

TEAM PROFILE

PROJECT SUMMARY, SCOPE
AND TARGET AUDIENCE

PROJECT OBJECTIVE
AND SOCIAL BENEFIT

DEFINITION AND SIGNIFICANCE
OF THE PROBLEM

PROPOSED SOLUTION

METHOD

PROTOTYPE / DESIGN
AND FUNCTIONING

COMPARISON WITH
EXISTING SOLUTIONS

COMMERCIALISATION
POTENTIAL AND SUSTAINABILITY

PROJECT TIMELINE

Inside the System & Real-World Use

KEY CAPABILITIES

- 1 **Mission Planner** Waypoint & geofence
- 2 **Live Video** + MAVLink telemetry
- 3 **Real-Time AI** on device 4 class YOLO detection (>15 FPS, F1 > 80%)
- 4 **Offline-First** works without internet



Geofencing health map

Plan

Autonomous survey over a 30 ha block

Fly & Scan

Live video classified by YOLO in real time

Flag

Stress/disease zones auto-pinpointed on the map

Act

Same-day treatment 4–6 hrs scouting becomes one flight

TEAM PROFILE

PROJECT SUMMARY, SCOPE AND TARGET AUDIENCE

PROJECT OBJECTIVE AND SOCIAL BENEFIT

DEFINITION AND SIGNIFICANCE OF THE PROBLEM

PROPOSED SOLUTION

METHOD

PROTOTYPE / DESIGN AND FUNCTIONING

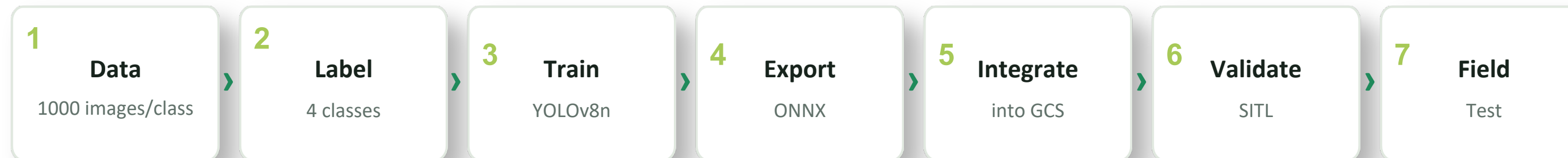
COMPARISON WITH EXISTING SOLUTIONS

COMMERCIALISATION POTENTIAL AND SUSTAINABILITY

PROJECT TIMELINE

Method & Workflow

DEVELOPMENT PIPELINE



Approach	Technologies & Tools	Analysis	Materials
Iterative develop-and-validate, using Software-in-the-Loop (SITL) simulation before real field tests.	Uno Platform (C#, one codebase → Win/Linux/Android), ONNX Runtime, OpenCV, MAVLink, ArduPilot / PX4 SITL.	Inference at 416×416 with frame-skipping & async pipeline to sustain >15 FPS; target detection F1 > 80%.	Low-cost UAV (\$300–800 class), companion laptop / tablet, telemetry radio.

TEAM PROFILE	
PROJECT SUMMARY, SCOPE AND TARGET AUDIENCE	
PROJECT OBJECTIVE AND SOCIAL BENEFIT	
DEFINITION AND SIGNIFICANCE OF THE PROBLEM	
PROPOSED SOLUTION	
METHOD	
PROTOTYPE / DESIGN AND FUNCTIONING	
COMPARISON WITH EXISTING SOLUTIONS	
COMMERCIALISATION POTENTIAL AND SUSTAINABILITY	
PROJECT TIMELINE	

Methodology & AI Validation

METHODOLOGY — DEVELOP & VALIDATE

- 1

Data & Augmentation
1000 image / class, 4 balanced classes, flip / scale / brightness augmentation
- 2

Model Training
YOLOv8n transfer learning , 200 epochs, 416x416 input resolution
- 3

Validation
Per-class confusion matrix on labeled validation set
- 4

SITL Simulation
ArduPilot / PX4 software-in-the-loop, mission & telemetry checks
- 5

Controlled Field Trials
Real low-cost UAV flights, accuracy benchmarked vs manual scouting

AI VALIDATION & RESULTS

≥ 0.85

Precision

≥ 0.82

Recall

> 0.80

F1-score

≥ 0.85

mAP@0.5

> 15

Inferences (FPS)

VALIDATION PROTOCOL
Per-class precision/recall/F1 from labeled validation set, field accuracy benchmarked against manual agronomist scouting, target false-positive rate <10%

Detection Class	Detections	Share	Avg Conf	Status
Lush Green	40,689	48.0%	0.75	✓ High
Inconsistent Growth	20,769	24.5%	0.68	✓ Moderate
Drought/Severe Stress	19,435	22.9%	0.71	✓ High
Bare Soil/Gap	3,848	4.5%	0.76	✓ High

Detection log — 20-min controlled validation run (recorded flight video): 12,000 frames · 84,741 detections · mean FHI 81.6.
Real field-flight benchmarking vs manual agronomist scouting is in progress (WP6).

Target Metriks - Validation Goals

TEAM PROFILE

PROJECT SUMMARY, SCOPE AND TARGET AUDIENCE

PROJECT OBJECTIVE AND SOCIAL BENEFIT

DEFINITION AND SIGNIFICANCE OF THE PROBLEM

PROPOSED SOLUTION

METHOD

PROTOTYPE / DESIGN AND FUNCTIONING

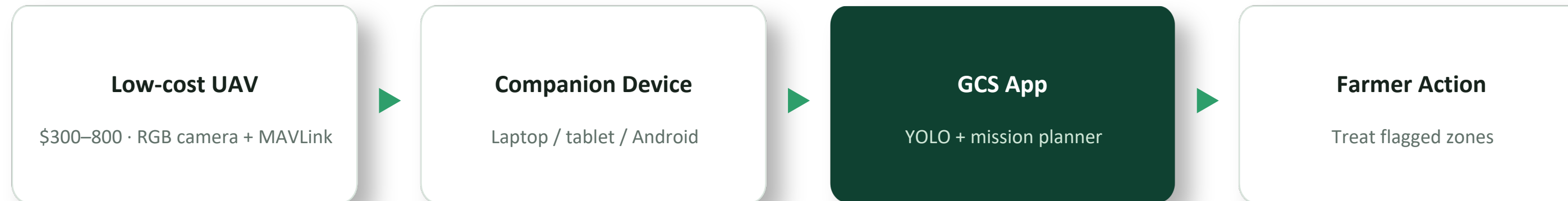
COMPARISON WITH EXISTING SOLUTIONS

COMMERCIALISATION POTENTIAL AND SUSTAINABILITY

PROJECT TIMELINE

Prototype Design & Architecture

SYSTEM ECOSYSTEM



TECHNOLOGY STACK

HARDWARE

- Low-cost UAV (\$300–800 class)
- RGB camera / live video feed
- Telemetry radio (MAVLink)
- Companion laptop / tablet / Android

SOFTWARE

- Uno Platform (C#, 1 codebase → 3 OS)
- ONNX Runtime (on-device YOLOv8n)
- OpenCV vegetation-health overlay
- MAVLink · ArduPilot / PX4 SITL

4-class detection · 416×416 inference · >15 FPS on-device · Fully offline (Edge Mode) · One C# codebase → 3 platforms

TEAM PROFILE

PROJECT SUMMARY, SCOPE
AND TARGET AUDIENCE

PROJECT OBJECTIVE
AND SOCIAL BENEFIT

DEFINITION AND SIGNIFICANCE
OF THE PROBLEM

PROPOSED SOLUTION

METHOD

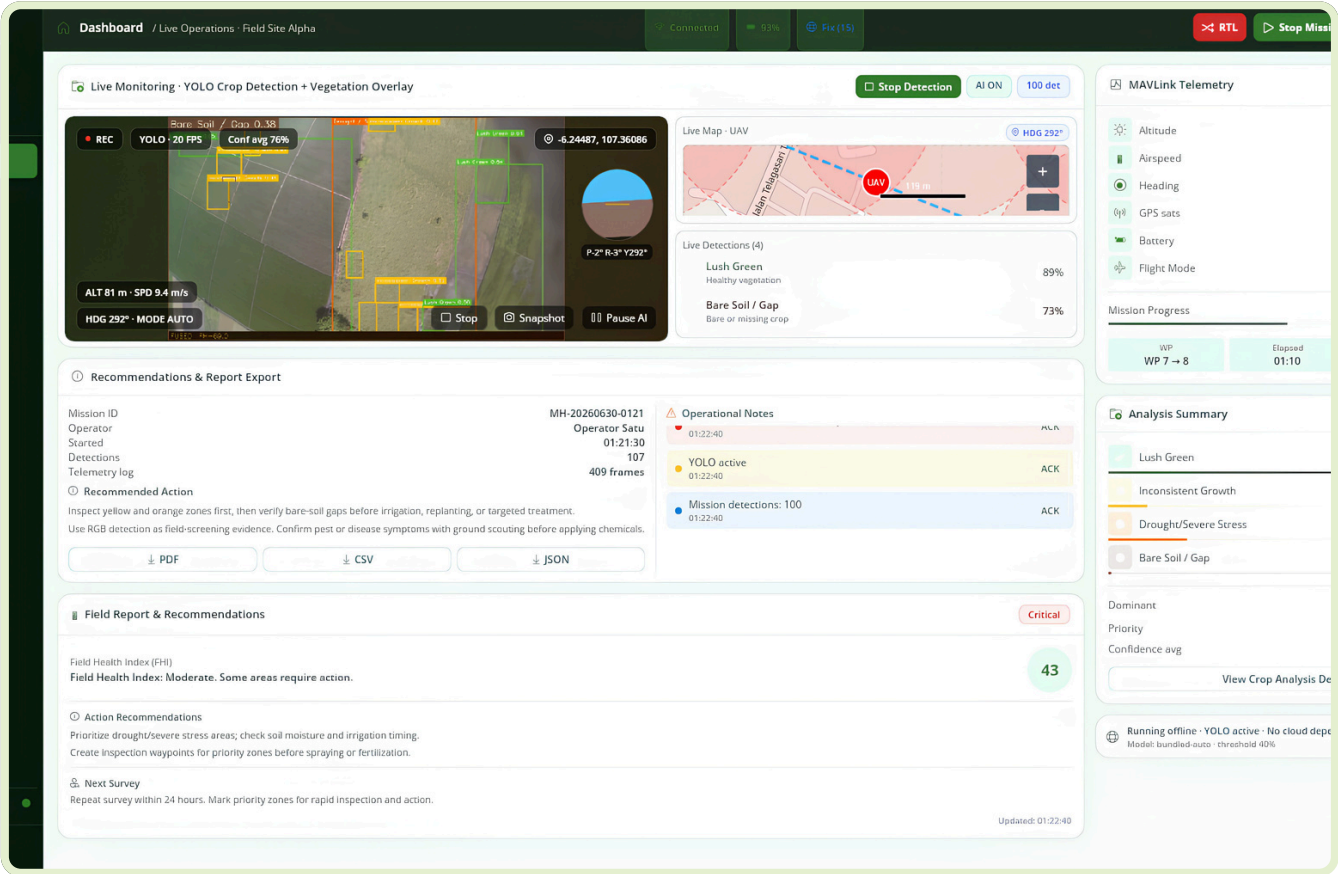
PROTOTYPE / DESIGN
AND FUNCTIONING

COMPARISON WITH
EXISTING SOLUTIONS

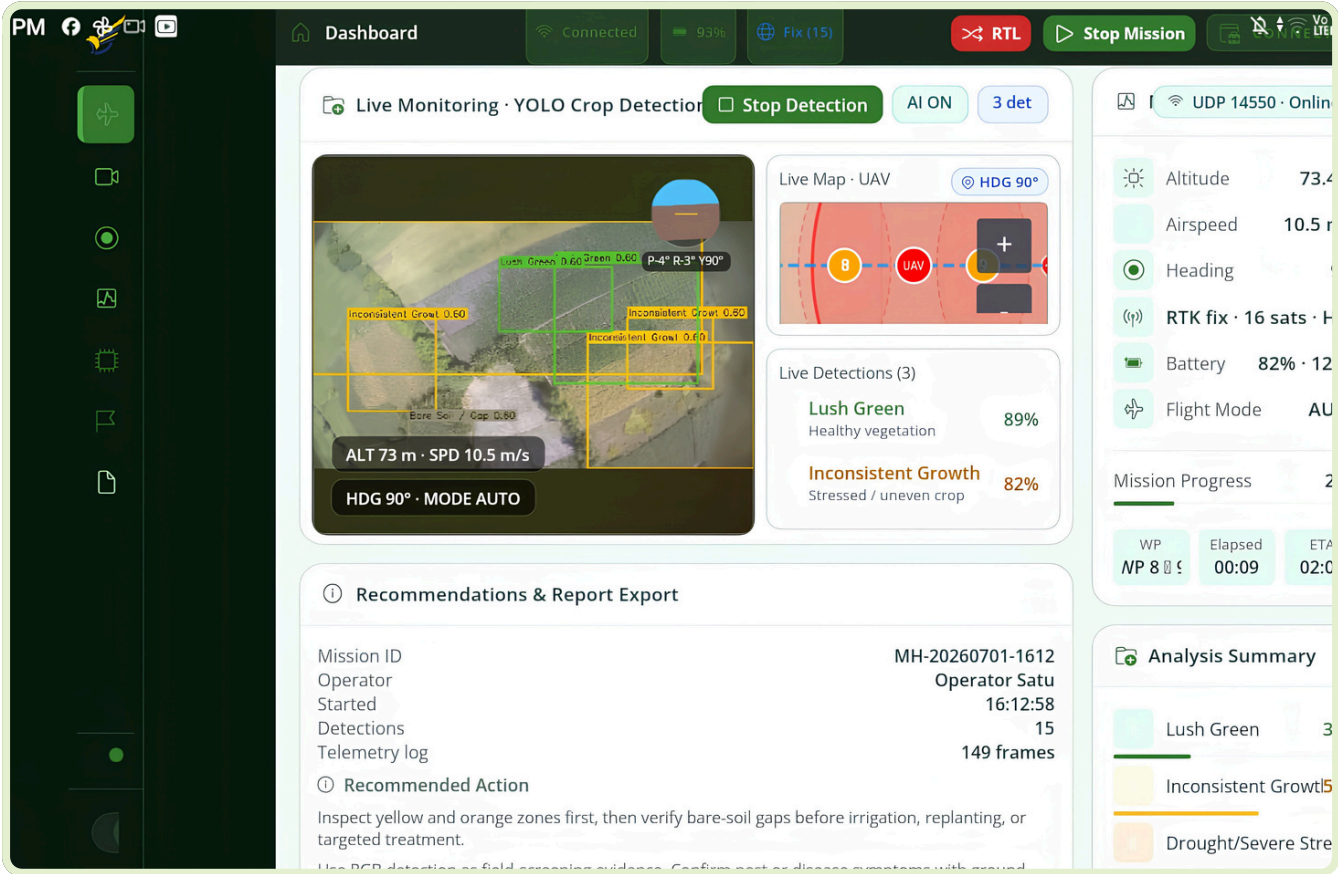
COMMERCIALISATION
POTENTIAL AND SUSTAINABILITY

PROJECT TIMELINE

Prototype Functioning & Evidence



Windows / Desktop



Android Tablet

Software build complete — live YOLO detection · vegetation overlay · MAVLink telemetry · waypoint & geofence · fully offline (Edge Mode).
Currently in field validation (WP6) — verifying real-flight accuracy & usability vs manual scouting.

TEAM PROFILE

PROJECT SUMMARY, SCOPE
AND TARGET AUDIENCE

PROJECT OBJECTIVE
AND SOCIAL BENEFIT

DEFINITION AND SIGNIFICANCE
OF THE PROBLEM

PROPOSED SOLUTION

METHOD

PROTOTYPE / DESIGN
AND FUNCTIONING

COMPARISON WITH
EXISTING SOLUTIONS

COMMERCIALISATION
POTENTIAL AND SUSTAINABILITY

PROJECT TIMELINE

Why MoonHarvest Is Different

Solution	Entry cost	Connectivity	Real-time AI	Hardware / best fit
MoonHarvest	\$300–800 + free app	Fully offline	On-device YOLO	Any MAVLink UAV · smallholders
DJI Agras / Terra	\$10k–30k	Desktop / cloud	Limited / cloud	Multispectral, vendor-locked · large farms
Pix4Dfields	Paid desktop license	Offline maps, no live AI	Post-process only	DJI & others · field mapping
DroneDeploy	Subscription	Cloud-based	Cloud analytics	Multi-drone · enterprise
EOS / Sentra	Sensor + subscription	Connectivity-dependent	Cloud / limited	Multispectral · large ops
Manual scouting	Low cash, high time	—	None	Human walk · very small plots

ORIGINAL CONTRIBUTIONS Zero-internet edge AI · One C# codebase → 3 platforms · Runtime-swappable YOLO models

Built for smallholder and cooperative use in emerging markets, especially areas with limited internet access and limited access to expensive agricultural drones.

TEAM PROFILE	
PROJECT SUMMARY, SCOPE AND TARGET AUDIENCE	
PROJECT OBJECTIVE AND SOCIAL BENEFIT	
DEFINITION AND SIGNIFICANCE OF THE PROBLEM	
PROPOSED SOLUTION	
METHOD	
PROTOTYPE / DESIGN AND FUNCTIONING	
COMPARISON WITH EXISTING SOLUTIONS	
COMMERCIALISATION POTENTIAL AND SUSTAINABILITY	
PROJECT TIMELINE	

Market & Value

USD 9.5B

Precision-ag software market
12.7% CAGR

10–40×

Cheaper entry
than commercial systems

3× lower

Maintenance cost
one codebase, open stack

Value Proposition

With low-cost UAV hardware and license-free core software, MoonHarvest lowers the entry barrier for precision agriculture by an estimated 10–40×.

Producible Now

Built on commodity hardware and mature open frameworks no exotic components required.

Partnerships

Agricultural cooperatives, agritech distributors, and university extension services.

Sustainability

Runtime-swappable AI keeps it current; cooperative distribution and recurring updates sustain value.

TEAM PROFILE

PROJECT SUMMARY, SCOPE
AND TARGET AUDIENCE

PROJECT OBJECTIVE
AND SOCIAL BENEFIT

DEFINITION AND SIGNIFICANCE
OF THE PROBLEM

PROPOSED SOLUTION

METHOD

PROTOTYPE / DESIGN
AND FUNCTIONING

COMPARISON WITH
EXISTING SOLUTIONS

COMMERCIALISATION
POTENTIAL AND SUSTAINABILITY

PROJECT TIMELINE

Business Model & Go-to-Market

BUSINESS MODEL

Open-core + Freemium

Free offline GCS app

Drives fast, low-friction adoption

Paid premium analytics

Recurring SaaS upsell on free base

Cooperative licensing

Annual contracts stable revenue

Low acquisition cost

Reach via co-ops & universities

10–40× cheaper entry vs DJI Agras

Payback < 1 growing season

REVENUE STREAMS

Premium Analytics

1

SaaS subscription

Mid-size farms & agronomists

Cooperative Licensing

2

Annual per-cooperative

Farmer co-ops & estates

Training & Support

3

Onboarding & service packs

Extension agencies

Custom AI Models

4

Per-crop model dev

Agritech & gov programs

GO-TO-MARKET

Pilot · 2027

Co-ops & demo farms validate value

Extend · 2028

University extension & training programs

Distribute · 2029

Regional agritech distributor partners

Scale · 2030

Premium SaaS tier & multi-crop models

TEAM PROFILE

PROJECT SUMMARY, SCOPE AND TARGET AUDIENCE

PROJECT OBJECTIVE AND SOCIAL BENEFIT

DEFINITION AND SIGNIFICANCE OF THE PROBLEM

PROPOSED SOLUTION

METHOD

PROTOTYPE / DESIGN AND FUNCTIONING

COMPARISON WITH EXISTING SOLUTIONS

COMMERCIALISATION POTENTIAL AND SUSTAINABILITY

PROJECT TIMELINE

Market USD 9.5B @ 12.7% CAGR · License-free open-source core keeps margins high and maintenance 3× lower · Year-1 target: 5 pilot cooperatives → 1,000+ ha monitored

Roadmap to TEKNOFEST 2026

Work Package No	WORK PACKAGE NAME	SUB-ACTIVITIES	START DATE	FINISH DATE
1	Concept & Requirements	Problem research, target users, requirements	Jan 2026	Feb 2026
2	Dataset & AI Model	Image collection, labeling, YOLOv8n training, ONNX export	Feb 2026	Apr 2026
3	GCS Core Development	Uno Platform app, video + telemetry, mission planner	Mar 2026	May 2026
4	AI Integration	On-device inference, OpenCV overlay, optimize >15 FPS	Apr 2026	May 2026
5	SITL Validation	Software-in-the-loop testing, F1 evaluation	May 2026	Jun 2026
6	Field Testing	Real UAV flights, accuracy & usability validation	Jun 2026	Jul 2026
7	Docs	Documentation	Jul 2026	Aug 2026

- TEAM PROFILE
- PROJECT SUMMARY, SCOPE AND TARGET AUDIENCE
- PROJECT OBJECTIVE AND SOCIAL BENEFIT
- DEFINITION AND SIGNIFICANCE OF THE PROBLEM
- PROPOSED SOLUTION
- METHOD
- PROTOTYPE / DESIGN AND FUNCTIONING
- COMPARISON WITH EXISTING SOLUTIONS
- COMMERCIALISATION POTENTIAL AND SUSTAINABILITY
- PROJECT TIMELINE

THANK YOU

Democratizing Precision Agriculture,
One Frame at a Time.

